Unit 2: Java Fundamental Concepts **7 hours**

2.1 Java Tokens, Keywords, Statements

Tokens are the basic building blocks of any code script. These include the keywords, operators, constants or other unique syntactic stuff specific to the language. The java compiler identifies these tokens based on delimiters and converts them into bytecode. Now, you must be wondering what these delimiters are and what bytecode is.

Delimiters
Most languages consider spaces, tabs, new lines, carriage returns and new feed-character as a delimiter. These delimiters are not part of the token but help identify tokens.

Bytecode
It is the object code that an interpreter converts into machine code that computers understand. These are compact numeric codes, references and constants that encode the result of compiler parsing.

Tokens are the smallest unit of the Java program. The compiler breaks down the lines of code into tokens. Let's see an example to understand tokens.
// program to understand java tokens

```
class HelloToken {
    public static void main(String[] args) {
        System.out.println("Hello BCA!!");
    }
}
```

In the above program, **"class, HelloToken, {, static, void, main, (, String, args, [,],), System, ., out, println"** etc. are tokens.

Types of Tokens
There are five types of java tokens: keywords, identifiers, literals, operators and separators. The classification is based on their work type; some are used to define names, and others for arithmetic operations.

1. **Keywords**
2. **Identifiers**
3. **Literals**
4. **Special Symbols**

Keywords
Keywords are reserved words in programming languages. These are used to indicate predefined terms and actions in a program. As a result, these words are not allowed to be used as names of variables or objects. It is case-sensitive and always written in lowercase. Java has the following keywords:

Abstract	**Assert**	**boolean**	**break**	**byte**
Case	Catch	char	class	continue
Default	do	double	else	enum
Extends	Final	finally	float	for
If	Implements	import	instanceof	int

Interface	Long	native	new	null
Package	Private	protected	public	return
Short	Static	strictfp	super	switch
Synchronized	This	throw	throws	transient
Try	Void	volatile	while	const/goto*

Indentifiers

A method name, class name, variable name, or label is an identifier in Java. The user typically defines these. The identifier names cannot be the same as any reserved keyword. Let's see an example to understand identifiers:

```
public class Test {
    public static void main(String[] args) {
        int num = 10;
    }
}
```

Identifiers present in the above program are:

Test: The name of the class.

main: The name of a method.

String: A predefined class name.

args: A variable name.

num: A variable name.

There are rules for naming identifiers.

- The characters allowed are **[A-Z], [a-z], [0-9], _ and $.**
- Identifiers are case-sensitive. That is, “ninja” is not the same as “NINJA”.
- Identifier names should not start with a digit. For example, “007IamNinja” is an invalid identifier.
- Whitespace is not allowed inside an identifier.
- Keywords can’t be used as an identifier.

Literals

Literals represent fixed values in a source code. These are similar to standard variables with the difference that these are constant. These can be classified as an integer literal, a string literal, a boolean etc. The user defines these mainly to define constants.

Syntax to define literals:

final data_type variable_name;

There are five types of literals in Java:

- **Integer**
- **Floating Point**
- **Boolean**
- **Character**
- **String**

Type	**Example**
Int	22
Double	25.08
Boolean	TRUE

Char	N
String	“Coding Ninjas”

Separators
These are the special symbols used to separate java tokens. These are sometimes called punctuators. The separators have special meaning and thus should not be used for anything else.

- **Brackets []:** These are used to define arrays and represent single and multi-dimension subscripts.
- **Braces {}:** These mark the start and end of multi-line code blocks.
- **Parenthesis ():** Used for function calls and parameters.
- **Comma (,):** These separate statements. For example, separate parameters.
- **Semicolon (;):** It invokes the initialization list.
- **Asterisk (*):** This is used to generate pointer variables.
- **Assignment operator (=):** It assigns values to variables. For example, a=10, here we are giving value 10 to variable 'a'.

Comments
In java there are multiple types of comments available in java:

1. **Single-line comments:** These are the comments that are denoted by // and are used to comment a single line of code.
2. **Multi-line comment:** These comments start with /* and end with */. When a user wants to comment on multiple lines in the code, these comments are used for it.
3. **Java doc comments**: These comments are denoted by "/**" and "*/". To provide documentation of Java code, these comments are used. Also, the Java doc tool is used to process these comments and generate HTML documents.

2.2 Constants and Variable

Constants

Java constants are variables whose value remains unchanged once they are assigned. A constant is declared using the “final” keyword, which means that its value cannot be changed. Generally, constants are declared at the top of the Java class with all other class variables.

The syntax to declare a constant is as follows:

static final datatype identifier_name=value;

For example, price is a variable that we want to make constant.

static final double PRICE=432.78;

Variables

The variable is the basic unit of storage in a Java program. A variable is defined by the combination of an identifier, a type, and an optional initializer.

Declaring a Variable

In Java, all variables must be declared before they can be used. The basic form of a variable declaration is shown here:

type identifier [= value][, identifier [= value] …];

Here, type is one of Java's atomic types, or the name of a class or interface.

The identifier is the name of the variable. You can initialize the variable by specifying an equal sign and a value. Keep in mind that the initialization expression must result in a value of the same (or compatible) type as that specified for the variable. To declare more than one variable of the specified type, use a comma-separated list.

Here are several examples of variable declarations of various types. Note that some include an initialization.

```
int a, b, c;           // declares three integers: a, b, and c
int d = 3, e, f = 5;   // declares three more integers, initializes d with 3 and
f with 5
byte z = 22;           // initializes z as a byte with value 22
double pi = 3.14159;   // declares a double variable pi with an approximation of
π
char x = 'x';          // declares a character variable x with value 'x'
```

Dynamic Initialization

Although the preceding examples have used only constants as initializers, Java allows variables to be initialized dynamically, using any expression valid at the time the variable is declared. For example, here is a short program that computes the length of the hypotenuse of a right triangle given the lengths of its two opposing sides:

```
// Demonstrate dynamic initialization.
class DynInit {
    public static void main(String args[]) {
        double a = 3.0, b = 4.0;

        // c is dynamically initialized
        double c = Math.sqrt(a * a + b * b);

        System.out.println("Hypotenuse is " + c);
```

```
    }
}
```

Here, three local variables—a, b, and c—are declared. The first two, a and b, are initialized by constants. However, c is initialized dynamically to the length of the hypotenuse (using the Pythagorean theorem).

2.3 Data Types

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Java Is a Strongly Typed Language

The Primitive Types

Java defines eight primitive types of data: byte, short, int, long, char, float, double, and boolean. The primitive types are also commonly referred to as simple types.

These can be put in four groups:

- Integers
  This group includes byte, short, int, and long, which are for whole-valued signed numbers.
- Floating-point numbers
  This group includes float and double, which represent numbers with fractional precision.
- Characters
  This group includes char, which represents symbols in a character set, like letters and numbers.
- Boolean
  This group includes boolean, which is a special type for representing true/false values.

Integers

Java defines four integer types: byte, short, int, and long. All of these are signed, positive and negative values. Java does not support unsigned, positive only integers.

The width and ranges of these integer types vary widely, as shown in this table

Name	Width	Range
long	64	–9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
int	32	–2,147,483,648 to 2,147,483,647
short	16	–32,768 to 32,767
byte	8	–128 to 127

Let’s look at each type of integer.

byte

The smallest integer type is byte. This is a signed 8-bit type that has a range from –128 to 127. Variables of type byte are especially useful when you’re working with a stream of data from a network or file.

Byte variables are declared by use of the byte keyword.

For example, the following declares two-byte variables called b and c:

byte b, c;

short

short is a signed 16-bit type. It has a range from –32,768 to 32,767. It is probably the least-used Java type. Here are some examples of short variable declarations:

short s;

short t;

int

The most commonly used integer type is int. It is a signed 32-bit type that has a range from –2,147,483,648 to 2,147,483,647. In addition to other uses, variables of type int are commonly employed to control loops and to index arrays.

long

long is a signed 64-bit type and is useful for those occasions where an int type is not large enough to hold the desired value. The range of a long is quite large.

For example, here is a program that computes the number of miles that light will travel in a specified number of days:

```
// Compute distance light travels using long variables.
class Light {
    public static void main(String args[]) {
        int lightspeed;
        long days;
        long seconds;
        long distance;

        // approximate speed of light in miles per second
        lightspeed = 186000;
```

```
        days = 1000;  // specify number of days here

        seconds = days * 24 * 60 * 60;
        // convert days to seconds

        distance = lightspeed * seconds;
        // compute distance: speed × time

        System.out.println("In " + days + " days light will travel about " +
distance + " miles.");
    }
}
```

This program generates the following output:

In 1000 days, light will travel about 16070400000000 miles.

Clearly, the result could not have been held in an int variable.

Floating-Point Types

Floating-point numbers, also known as real numbers, are used when evaluating expressions that require fractional precision.

There are two kinds of floating-point types, float and double, which represent single- and double-precision numbers, respectively. Their width and ranges are shown here:

Name	Width in Bits	Approximate Range
double	64	4.9e–324 to 1.8e+308
float	32	1.4e–045 to 3.4e+038

Each of these floating-point types is examined next.

float

The type float specifies a single-precision value that uses 32 bits of storage. Single precision is faster on some processors and takes half as much space as double precision, but will become imprecise when the values are either very large or very small. Variables of type float are useful when you need a fractional component, but don't require a large degree of precision. For example, float can be useful when representing dollars and cents. Here are some example float variable declarations:

float highTemp, lowTemp;

double

Double precision, as denoted by the double keyword, uses 64 bits to store a value. Double precision is actually faster than single precision on some modern processors that have been optimized for high-

speed mathematical calculations. All transcendental math functions, such as sin( ), cos( ), and sqrt( ), return double values. When you need to maintain accuracy over many iterative calculations, or are manipulating large-valued numbers, double is the best choice.

Here is a short program that uses double variables to compute the area of a circle:

```
// Compute the area of a circle.
class Area {
    public static void main(String args[]) {
        double pi, r, a;

        r = 10.8; // radius of circle
        pi = 3.1416; // pi, approximately
        a = pi * r * r; // compute area

        System.out.println("Area of circle is " + a);
    }
}
```

Characters

In Java, the data type used to store characters is char. A key point to understand is that Java uses Unicode to represent characters. Unicode defines a fully international character set that can represent all of the characters found in all human languages. It is a unification of dozens of character sets, such as Latin, Greek, Arabic, Cyrillic, Hebrew, Katakana, Hangul, and many more. At the time of Java's creation, Unicode required 16 bits. Thus, in Java char is a 16-bit type. The range of a char is 0 to 65,536. There are no negative chars. The standard set of characters known as ASCII still ranges from 0 to 127 as always, and the extended 8-bit character set, ISO-Latin-1, ranges from 0 to 255. Since Java is designed to allow programs to be written for worldwide use, it makes sense that it would use Unicode to represent characters. Of course, the use of Unicode is somewhat inefficient for languages such as English, German, Spanish, or French, whose characters can easily be contained within 8 bits. But such is the price that must be paid for global portability.

Here is a program that demonstrates char variables:

```
// Demonstrate char data type.
class CharDemo {
    public static void main(String args[]) {
        char ch1, ch2;

        ch1 = 88; // code for X
        ch2 = 'Y';

        System.out.print("ch1 and ch2: ");
        System.out.println(ch1 + " " + ch2);
    }
}
```

This program displays the following output:

ch1 and ch2: X Y

Another example, you can add two characters together, or increment the value of a character variable. Consider the following program:

```
// char variables behave like integers.
class CharDemo2 {
    public static void main(String args[]) {
        char ch1;

        ch1 = 'X';
        System.out.println("ch1 contains " + ch1);

        ch1++; // increment ch1
        System.out.println("ch1 is now " + ch1);
    }
}
```

The output generated by this program is shown here:

ch1 contains X

ch1 is now Y

Booleans

Java has a primitive type, called boolean, for logical values. It can have only one of two possible values, true or false.

Here is a program that demonstrates the boolean type:

```
// Demonstrate boolean values.
class BoolTest {
    public static void main(String args[]) {
        boolean b;

        b = false;
        System.out.println("b is " + b);
        b = true;
        System.out.println("b is " + b);

        // a boolean value can control the if statement
        if(b) System.out.println("This is executed.");

        b = false;
        if(b) System.out.println("This is not executed.");

        // outcome of a relational operator is a boolean value
        System.out.println("10 > 9 is " + (10 > 9));
    }
}
```

The output generated by this program is shown here:

b is false

b is true

This is executed.

10 > 9 is true

2.4 Operators

Operators

Java provides a rich operator environment. Most of its operators can be divided into the following four groups: arithmetic, bitwise, relational, and logical.

Arithmetic Operators

Arithmetic operators are used in mathematical expressions in the same way that they are used in algebra. The following table lists the arithmetic operators:

Operator	Result
+	Addition (also unary plus)
–	Subtraction (also unary minus)
*	Multiplication
/	Division
%	Modulus
++	Increment
+=	Addition assignment
– =	Subtraction assignment
*=	Multiplication assignment
/=	Division assignment
%=	Modulus assignment
– –	Decrement

The operands of the arithmetic operators must be of a numeric type. You cannot use them on boolean types, but you can use them on char types, since the char type in Java is, essentially, a subset of int.

The Basic Arithmetic Operators

The basic arithmetic operations—addition, subtraction, multiplication, and division—all behave as you would expect for all numeric types.

The following simple example program demonstrates the arithmetic operators. It also illustrates the difference between floating-point division and integer division.

```
// Demonstrate the basic arithmetic operators.
class BasicMath {
    public static void main(String args[]) {
        // arithmetic using integers
        System.out.println("Integer Arithmetic");
        int a = 1 + 1;
        int b = a * 3;
        int c = b / 4;
        int d = c - a;
        int e = -d;

        System.out.println("a = " + a);
        System.out.println("b = " + b);
        System.out.println("c = " + c);
        System.out.println("d = " + d);
        System.out.println("e = " + e);

        // arithmetic using doubles
```

```
        System.out.println("Floating Point Arithmetic");
        double da = 1 + 1;
        double db = da * 3;
        double dc = db / 4;
        double dd = dc - da;
        double de = -dd;
        System.out.println("da = " + da);
        System.out.println("db = " + db);
        System.out.println("dc = " + dc);
        System.out.println("dd = " + dd);
        System.out.println("de = " + de);
    }
}
```

When you run this program, you will see the following output:

Integer Arithmetic

a = 2

b = 6

c = 1

d = -1

e = 1

Floating Point Arithmetic

da = 2.0

db = 6.0

dc = 1.5

dd = -0.5

de = 0.5

The Modulus Operator

The modulus operator, %, returns the remainder of a division operation. It can be applied to floating-point types as well as integer types. The following example program demonstrates the %:

```
// Demonstrate the % operator.
class Modulus {
    public static void main(String args[]) {
        int x = 42;
        double y = 42.25;

        System.out.println("x mod 10 = " + x % 10);
        System.out.println("y mod 10 = " + y % 10);
    }
}
```

When you run this program, you will get the following output:

x mod 10 = 2

y mod 10 = 2.25

Arithmetic Compound Assignment Operators

Java provides special operators that can be used to combine an arithmetic operation with an assignment. As you probably know, statements like the following are quite common in programming:

a = a + 4;

In Java, you can rewrite this statement as shown here:

a += 4;

This version uses the += compound assignment operator. Both statements perform the same action: they increase the value of a by 4.

Here is another example,

a = a % 2;

which can be expressed as

a %= 2;

In this case, the %= obtains the remainder of a /2 and puts that result back into a.

Here is a sample program that shows several op= assignments in action:

```
// Demonstrate several assignment operators.
class OpEquals {
    public static void main(String args[]) {
        int a = 1;
        int b = 2;
        int c = 3;

        a += 5;
        b *= 4;
        c += a * b;
        c %= 6;

        System.out.println("a = " + a);
        System.out.println("b = " + b);
        System.out.println("c = " + c);
    }
}
```

The output of this program is shown here:

a = 6

b = 8

c = 3

Increment and Decrement

The ++ and the – – are Java's increment and decrement operators. The increment operator increases its operand by one. The decrement operator decreases its operand by one.

For example, this statement: x = x + 1; can be rewritten like this by use of the increment operator: x++;

Similarly, this statement: x = x - 1; is equivalent to x--;

These operators are unique in that they can appear both in postfix form, where they follow the operand as just shown, and prefix form, where they precede the operand.

In the prefix form, the operand is incremented or decremented before the value is obtained for use in the expression. In postfix form, the previous value is obtained for use in the expression, and then the operand is modified.

For example:

x = 42;

y = ++x;

In this case, y is set to 43 as you would expect, because the increment occurs before x is assigned to y. Thus, the line y = ++x; is the equivalent of these two statements:

x = x + 1;

y = x;

However, when written like this,

x = 42;

y = x++;

the value of x is obtained before the increment operator is executed, so the value of y is 42.

Of course, in both cases x is set to 43. Here, the line y = x++; is the equivalent of these two statements:

y = x;

x = x + 1;

The following program demonstrates the increment operator.

```
// Demonstrate ++.
class IncDec {
    public static void main(String args[]) {
```

```
        int a = 1;
        int b = 2;
        int c;
        int d;

        c = ++b;
        d = a++;

        c++;

        System.out.println("a = " + a);
        System.out.println("b = " + b);
        System.out.println("c = " + c);
        System.out.println("d = " + d);
    }
}
```

The output of this program follows:

a = 2

b = 3

c = 4

d = 1

The Bitwise Operators

Java defines several bitwise operators that can be applied to the integer types: long, int, short, char, and byte. These operators act upon the individual bits of their operands. They are summarized in the following table:

Operator	Result
~	Bitwise unary NOT
&	Bitwise AND
|	Bitwise OR
^	Bitwise exclusive OR
>>	Shift right
>>>	Shift right zero fill
<<	Shift left
&=	Bitwise AND assignment
|=	Bitwise OR assignment
^=	Bitwise exclusive OR assignment
>>=	Shift right assignment
>>>=	Shift right zero fill assignment
<<=	Shift left assignment

Relational Operators

The relational operators determine the relationship that one operand has to the other. Specifically, they determine equality and ordering. The relational operators are shown here:

Operator	Result
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

The outcome of these operations is a boolean value. The relational operators are most frequently used in the expressions that control the if statement and the various loop statements.

As stated, the result produced by a relational operator is a boolean value. For example, the following code fragment is perfectly valid:

```
int a = 4;

int b = 1;

boolean c = a < b;
```

In this case, the result of a<b (which is false) is stored in c.

Boolean Logical Operators

The Boolean logical operators shown here operate only on boolean operands. All of the binary logical operators combine two boolean values to form a resultant boolean value.

Operator	Result
&	Logical AND
|	Logical OR
^	Logical XOR (exclusive OR)
||	Short-circuit OR
&&	Short-circuit AND
!	Logical unary NOT
&=	AND assignment
|=	OR assignment
^=	XOR assignment
==	Equal to
!=	Not equal to
?:	Ternary if-then-else

The logical Boolean operators, &, |, and ^, operate on boolean values in the same way that they operate on the bits of an integer. The logical ! operator inverts the Boolean state: !true == false and !false == true. The following table shows the effect of each logical operation:

A	B	A | B	A & B	A ^ B	!A
False	False	False	False	False	True
True	False	True	False	True	False
False	True	True	False	True	True
True	True	True	True	False	False

Here is a program that is almost the same as the BitLogic example shown earlier, but it operates on boolean logical values instead of binary bits:

```
// Demonstrate the boolean logical operators.
class BoolLogic {
    public static void main(String args[]) {
        boolean a = true;
        boolean b = false;
        boolean c = a | b;
        boolean d = a & b;
        boolean e = a ^ b;
        boolean f = (a & b) | (a & !b);
        boolean g = !a;

        System.out.println("    a = " + a);
        System.out.println("    b = " + b);
        System.out.println("    a|b = " + c);
        System.out.println("    a&b = " + d);
        System.out.println("    a^b = " + e);
        System.out.println("    !a&b|a&!b = " + f);
        System.out.println("    !a = " + g);
    }
}
```

After running this program, you will see that the same logical rules apply to boolean values as they did to bits. As you can see from the following output, the string representation of a Java boolean value is one of the literal values true or false:

The content shows the results of boolean logical operations with a = true and b = false:

- a | b = true (OR operation)
- a & b = false (AND operation)
- a ^ b = true (XOR operation)
- !a & b | a & !b = true (combined operation)
- !a = false (NOT operation)

Short-Circuit Logical Operators

Java provides two interesting Boolean operators not found in some other computer languages. These are secondary versions of the Boolean AND and OR operators, and are commonly known as short-circuit logical operators.

As you can see from the preceding table, the OR operator results in true when A is true, no matter what B is. Similarly, the AND operator results in false when A is false, no matter what B is. If you use the || and && forms, rather than the | and & forms of these operators, Java will not bother to evaluate the right-hand operand when the outcome of the expression can be determined by the left operand alone. This is very useful when the right-hand operand depends on the value of the left one in order to function properly. For example, the following code fragment shows how you can take advantage of short-circuit logical evaluation to be sure that a division operation will be valid before evaluating it:

```
if (denom != 0 && num / denom > 10)
```

Since the short-circuit form of AND (&&) is used, there is no risk of causing a run-time exception when denom is zero. If this line of code were written using the single & version of AND, both sides would be evaluated, causing a run-time exception when denom is zero. It is standard practice to use the short-circuit forms of AND and OR in cases involving Boolean logic, leaving the single-character versions exclusively for bitwise operations. However, there are exceptions to this rule. For example, consider the following statement:

```
if(c==1 & e++ < 100) d = 100;
```

Here, using a single & ensures that the increment operation will be applied to e whether c is equal to 1 or not.

The Assignment Operator

The assignment operator is the single equal sign, =. The assignment operator works in Java much as it does in any other computer language. It has this general form: var = expression; Here, the type of var must be compatible with the type of expression. The assignment operator does have one interesting attribute that you may not be familiar with: it allows you to create a chain of assignments. For example, consider this fragment:

```
int x, y, z;

x = y = z = 100; // set x, y, and z to 100
```

This fragment sets the variables x, y, and z to 100 using a single statement. This works because the = is an operator that yields the value of the right-hand expression. Thus, the value of z = 100 is 100, which is then assigned to y, which in turn is assigned to x. Using a “chain of assignment” is an easy way to set a group of variables to a common value.

The ? Operator

Java includes a special ternary (three-way) operator that can replace certain types of if-then-else statements. This operator is the ?. It can seem somewhat confusing at first, but the ? can be used very effectively once mastered. The ? has this general form:

```
expression1 ? expression2 : expression3
```

Here, expression1 can be any expression that evaluates to a boolean value. If expression1 is true, then expression2 is evaluated; otherwise, expression3 is evaluated. The result of the ? operation is that of

the expression evaluated. Both expression2 and expression3 are required to return the same (or compatible) type, which can't be void.

Here is an example of the way that the ? is employed:

ratio = denom == 0 ? 0 : num / denom;

When Java evaluates this assignment expression, it first looks at the expression to the left of the question mark. If denom equals zero, then the expression between the question mark and the colon is evaluated and used as the value of the entire ? expression. If denom does not equal zero, then the expression after the colon is evaluated and used for the value of the entire ? expression. The result produced by the ? operator is then assigned to ratio. Here is a program that demonstrates the ? operator. It uses it to obtain the absolute value of a variable.

```
// Demonstrate ?.
class Ternary {
    public static void main(String args[]) {
        int i, k;

        i = 10;
        k = i < 0 ? -i : i; // get absolute value of i
        System.out.print("Absolute value of ");
        System.out.println(i + " is " + k);

        i = -10;
        k = i < 0 ? -i : i; // get absolute value of i
        System.out.print("Absolute value of ");
        System.out.println(i + " is " + k);
    }
}
```

The output generated by the program is shown here:

Absolute value of 10 is 10

Absolute value of -10 is 10

2.5 Control Statements

A programming language uses control statements to cause the flow of execution to advance and branch based on changes to the state of a program. Java's program control statements can be put into the following categories: selection, iteration, and jump. Selection statements allow your program to choose different paths of execution based upon the outcome of an expression or the state of a variable. Iteration statements enable program execution to repeat one or more statements (that is, iteration statements form loops). Jump statements allow your program to execute in a nonlinear fashion. All of Java's control statements are examined here.

2.5.1 Branching
2.5.2 Looping Statements
2.5.3 Jump Statements

2.6 Arrays in Java

2.6.1 One dimensional
2.6.2 Two dimensional

Arrays
An array is a group of like-typed variables that are referred to by a common name. Arrays of any type can be created and may have one or more dimensions. A specific element in an array is accessed by its index. Arrays offer a convenient means of grouping related information.

One-Dimensional Arrays
A one-dimensional array is, essentially, a list of like-typed variables. To create an array, you first must create an array variable of the desired type. The general form of a one-dimensional array declaration is

type var-name[ ];

Here, type declares the element type (also called the base type) of the array. The element type determines the data type of each element that comprises the array. Thus, the element type for the array determines what type of data the array will hold. For example, the following declares an array named month_days with the type "array of int":

int month_days[];

Although this declaration establishes the fact that month_days is an array variable, no array actually exists. To link month_days with an actual, physical array of integers, you must allocate one using new and assign it to month_days. new is a special operator that allocates memory.
You will look more closely at new in a later chapter, but you need to use it now to allocate memory for arrays. The general form of new as it applies to onedimensional arrays appears as follows:

array-var = new type [size];

Here, type specifies the type of data being allocated, size specifies the number of elements in the array, and array-var is the array variable that is linked to the array. That is, to use new to allocate an array, you must specify the type and number of elements to allocate. The elements in the array allocated by new will automatically be initialized to zero (for numeric types), false (for boolean), or null (for reference types, which are described in a later chapter). This example allocates a 12-element array of integers and links them to month_days:

month_days = new int[12];

After this statement executes, month_days will refer to an array of 12 integers. Further, all elements in the array will be initialized to zero.
Let's review: Obtaining an array is a two-step process. First, you must declare a variable of the desired array type. Second, you must allocate the memory that will hold the array, using new, and assign it to the array variable. Thus, in Java all arrays are dynamically allocated. If the concept of dynamic allocation is unfamiliar to you, don't worry. It will be described at length later in this book.

Once you have allocated an array, you can access a specific element in the array by specifying its index within square brackets. All array indexes start at zero. For example, this statement assigns the value 28 to the second element of month_days:

month_days[1] = 28;

The next line displays the value stored at index 3:
System.out.println(month_days[3]);

Putting together all the pieces, here is a program that creates an array of the number of days in each month:

```
// Demonstrate a one-dimensional array.
class Array {
    public static void main(String args[]) {
        int month_days[];
        month_days = new int[12];

        month_days[0] = 31;
        month_days[1] = 28;
        month_days[2] = 31;
        month_days[3] = 30;
        month_days[4] = 31;
        month_days[5] = 30;
        month_days[6] = 31;
        month_days[7] = 31;
        month_days[8] = 30;
        month_days[9] = 31;
        month_days[10] = 30;
        month_days[11] = 31;

        System.out.println("April has " + month_days[3] + " days.");
    }
}
```

When you run this program, it prints the number of days in April. As mentioned, Java array indexes start with zero, so the number of days in April is month_days[3] or 30.
It is possible to combine the declaration of the array variable with the allocation of the array itself, as shown here:

int month_days[] = new int[12];

This is the way that you will normally see it done in professionally written Java programs.
Arrays can be initialized when they are declared. The process is much the same as that used to initialize the simple types. An array initializer is a list of comma-separated expressions surrounded by curly braces. The commas separate the values of the array elements. The array will automatically be created large enough to hold the number of elements you specify in the array initializer. There is no need to use new. For example, to store the number of days in each month, the following code creates an initialized array of integers:

```
// An improved version of the previous program.
class AutoArray {
    public static void main(String args[]) {
        int month_days[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

        System.out.println("April has " + month_days[3] + " days.");
    }
}
```

When you run this program, you see the same output as that generated by the previous version.
Java strictly checks to make sure you do not accidentally try to store or reference values outside of the range of the array. The Java run-time system will check that all array indexes are in the correct range. For example, the run-time system will check the value of each index into month_days to make sure that it is between 0 and 11 inclusive. If you try to access elements outside the range of the array (negative numbers or numbers greater than the length of the array), you will cause a run-time error.
Here is one more example that uses a one-dimensional array. It finds the average of a set of numbers.

```
// Average an array of values.
class Average {
    public static void main(String args[]) {
        double nums[] = {10.1, 11.2, 12.3, 13.4, 14.5};
        double result = 0;
        int i;

        for(i=0; i<5; i++)
            result = result + nums[i];

        System.out.println("Average is " + result / 5);
    }
}
```

Multidimensional Arrays
In Java, multidimensional arrays are implemented as arrays of arrays. To declare a multidimensional array variable, specify each additional index using another set of square brackets. For example, the following declares a two-dimensional array variable called twoD:

int twoD[][] = new int[4][5];

This allocates a 4 by 5 array and assigns it to twoD. Internally, this matrix is implemented as an array of arrays of int. Conceptually, this array will look like the one shown in Figure 3-1.

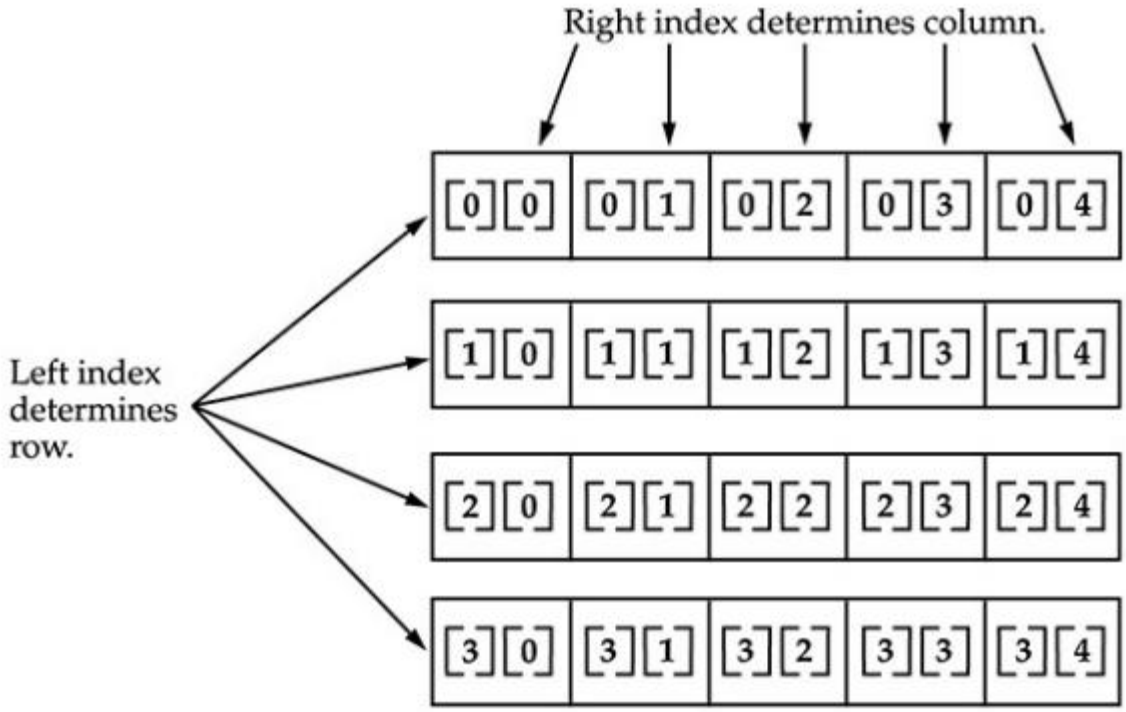


Figure 3-1 A conceptual view of a 4 by 5, two-dimensional array

The following program numbers each element in the array from left to right, top to bottom, and then displays these values:

```
// Demonstrate a two-dimensional array.
class TwoDArray {
  public static void main(String args[]) {
    int twoD[][]= new int[4][5];
    int i, j, k = 0;

    for(i=0; i<4; i++)
      for(j=0; j<5; j++) {
        twoD[i][j] = k;
        k++;
      }

    for(i=0; i<4; i++) {
      for(j=0; j<5; j++)
        System.out.print(twoD[i][j] + " ");
      System.out.println();
    }
  }
}
```

This program generates the following output:
0 1 2 3 4
5 6 7 8 9
10 11 12 13 14
15 16 17 18 19

When you allocate memory for a multidimensional array, you need only specify the memory for the first (leftmost) dimension. You can allocate the remaining dimensions separately. For example, this following code allocates memory for the first dimension of twoD when it is declared. It allocates the second dimension separately.

```
int twoD[][] = new int[4][];
twoD[0] = new int[5];
twoD[1] = new int[5];
twoD[2] = new int[5];
twoD[3] = new int[5];
```

While there is no advantage to individually allocating the second dimension arrays in this situation, there may be in others. For example, when you allocate dimensions individually, you do not need to allocate the same number of elements for each dimension. As stated earlier, since multidimensional arrays are actually arrays of arrays, the length of each array is under your control. For example, the following program creates a two-dimensional array in which the sizes of the second dimension are unequal:

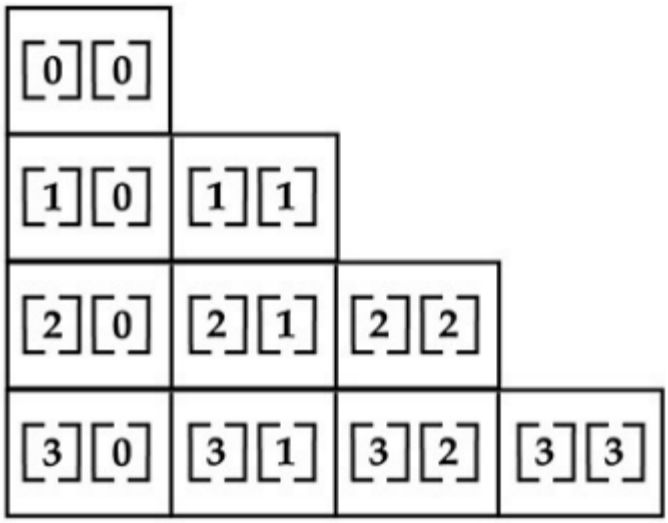
```
// Manually allocate differing size second dimensions.
class TwoDAgain {
  public static void main(String args[]) {
    int twoD[][] = new int[4][];
    twoD[0] = new int[1];
    twoD[1] = new int[2];
    twoD[2] = new int[3];
    twoD[3] = new int[4];

    int i, j, k = 0;

    for(i=0; i<4; i++)
      for(j=0; j<i+1; j++) {
      twoD[i][j] = k;
      k++;
    }

    for(i=0; i<4; i++) {
      for(j=0; j<i+1; j++)
        System.out.print(twoD[i][j] + " ");
      System.out.println();
    }
  }
}
```

This program generates the following output:
0
1 2
3 4 5
6 7 8 9
The array created by this program looks like this:



The use of uneven (or irregular) multidimensional arrays may not be appropriate for many applications, because it runs contrary to what people expect to find when a multidimensional array is encountered. However, irregular arrays can be used effectively in some situations. For example, if you need a very large two-dimensional array that is sparsely populated (that is, one in which not all of the elements will be used), then an irregular array might be a perfect solution.

It is possible to initialize multidimensional arrays. To do so, simply enclose each dimension's initializer within its own set of curly braces. The following program creates a matrix where each element contains the product of the row and column indexes. Also notice that you can use expressions as well as literal values inside of array initializers.

```
// Initialize a two-dimensional array.
class Matrix {
  public static void main(String args[]) {
    double m[][] = {
      { 0*0, 1*0, 2*0, 3*0 },
      { 0*1, 1*1, 2*1, 3*1 },
      { 0*2, 1*2, 2*2, 3*2 },
      { 0*3, 1*3, 2*3, 3*3 }
    };
    int i, j;

    for(i=0; i<4; i++) {
      for(j=0; j<4; j++)
        System.out.print(m[i][j] + " ");
      System.out.println();
    }
  }
}
```

When you run this program, you will get the following output:
0.0 0.0 0.0 0.0
0.0 1.0 2.0 3.0
0.0 2.0 4.0 6.0
0.0 3.0 6.0 9.0

As you can see, each row in the array is initialized as specified in the initialization lists.
Let's look at one more example that uses a multidimensional array. The following program creates a 3 by 4 by 5, three-dimensional array. It then loads each element with the product of its indexes. Finally, it displays these products.

```
// Demonstrate a three-dimensional array.
class ThreeDMatrix {
  public static void main(String args[]) {
    int threeD[][][] = new int[3][4][5];
    int i, j, k;

    for(i=0; i<3; i++)
      for(j=0; j<4; j++)
        for(k=0; k<5; k++)
          threeD[i][j][k] = i * j * k;

    for(i=0; i<3; i++) {
      for(j=0; j<4; j++) {
        for(k=0; k<5; k++)
          System.out.print(threeD[i][j][k] + " ");
        System.out.println();
      }
      System.out.println();
    }
  }
}
```

This program generates the following output:
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0

0 0 0 0 0
0 1 2 3 4
0 2 4 6 8
0 3 6 9 12

0 0 0 0 0
0 2 4 6 8
0 4 8 12 16
0 6 12 18 24

Alternative Array Declaration Syntax
There is a second form that may be used to declare an array:
type[ ] var-name;
Here, the square brackets follow the type specifier, and not the name of the array variable. For example, the following two declarations are equivalent:
int al[] = new int[3];
int[] a2 = new int[3];

The following declarations are also equivalent:
char twod1[][] = new char[3][4];
char[][] twod2 = new char[3][4];
This alternative declaration form offers convenience when declaring several arrays at the same time. For example,
int[] nums, nums2, nums3; // create three arrays
creates three array variables of type int. It is the same as writing
int nums[], nums2[], nums3[]; // create three arrays
The alternative declaration form is also useful when specifying an array as a return type for a method. Both forms are used in this book.

2.7 Java Strings

2.7.1 String Length
2.7.2 Concatenation
2.7.3 String Comparison

String Handling

A brief overview of Java's string handling was presented in Chapter 7. In this chapter, it is described in detail. As is the case in most other programming languages, in Java a string is a sequence of characters. But, unlike some other languages that implement strings as character arrays, Java implements strings as objects of type **String**.

Implementing strings as built-in objects allows Java to provide a full complement of features that make string handling convenient. For example, Java has methods to compare two strings, search for a substring, concatenate two strings, and change the case of letters within a string. Also, **String** objects can be constructed a number of ways, making it easy to obtain a string when needed.

Somewhat unexpectedly, when you create a **String** object, you are creating a string that cannot be changed. That is, once a **String** object has been created, you cannot change the characters that comprise that string. At first, this may seem to be a serious restriction. However, such is not the case. You can still perform all types of string operations. The difference is that each time you need an altered version of an existing string, a new **String** object is created that contains the modifications. The original string is left unchanged. This approach is used because fixed, immutable strings can be implemented more efficiently than changeable ones. For those cases in which a modifiable string is desired, Java provides two options: **StringBuffer** and **StringBuilder**. Both hold strings that can be modified after they are created.

The **String**, **StringBuffer**, and **StringBuilder** classes are defined in **java.lang**. Thus, they are available to all programs automatically. All are

declared **final**, which means that none of these classes may be subclassed. This allows certain optimizations that increase performance to take place on common string operations. All three implement the **CharSequence** interface.

One last point: To say that the strings within objects of type **String** are unchangeable means that the contents of the **String** instance cannot be changed after it has been created. However, a variable declared as a **String** reference can be changed to point at some other **String** object at any time.

The String Constructors

The **String** class supports several constructors. To create an empty **String**, call the default constructor. For example,

```
String s = new String();
```

will create an instance of **String** with no characters in it.

Frequently, you will want to create strings that have initial values. The **String** class provides a variety of constructors to handle this. To create a **String** initialized by an array of characters, use the constructor shown here:

String(char *chars*[ ])

Here is an example:

```
char chars[] = { 'a', 'b', 'c' };
String s = new String(chars);
```

This constructor initializes **s** with the string "abc".

You can specify a subrange of a character array as an initializer using the following constructor:

String(char *chars*[ ], int *startIndex*, int *numChars*)

Here, *startIndex* specifies the index at which the subrange begins, and *numChars* specifies the number of characters to use. Here is an example:

```
char chars[] = { 'a', 'b', 'c', 'd', 'e', 'f' };
String s = new String(chars, 2, 3);
```

This initializes **s** with the characters **cde**.

You can construct a **String** object that contains the same character sequence as another **String** object using this constructor:

String(String *strObj*)

Here, *strObj* is a **String** object. Consider this example:

```
// Construct one String from another.
class MakeString {
  public static void main(String args[]) {
    char c[] = {'J', 'a', 'v', 'a'};
    String s1 = new String(c);
    String s2 = new String(s1);

    System.out.println(s1);
    System.out.println(s2);
  }
}
```

The output from this program is as follows:

```
Java
Java
```

As you can see, **s1** and **s2** contain the same string.

Even though Java's **char** type uses 16 bits to represent the basic Unicode character set, the typical format for strings on the Internet uses arrays of 8-bit bytes constructed from the ASCII character set. Because 8-bit ASCII strings are common, the **String** class provides constructors that initialize a string when given a **byte** array. Two forms are shown here:

String(byte *chrs*[ ])
String(byte *chrs*[ ], int *startIndex*, int *numChars*)

Here, *chrs* specifies the array of bytes. The second form allows you to specify a subrange. In each of these constructors, the byte-to-character conversion is done by using the default character encoding of the platform. The following program illustrates these constructors:

```
// Construct string from subset of char array.
class SubStringCons {
  public static void main(String args[]) {
    byte ascii[] = {65, 66, 67, 68, 69, 70 };

    String s1 = new String(ascii);
    System.out.println(s1);

    String s2 = new String(ascii, 2, 3);
    System.out.println(s2);
  }
}
```

This program generates the following output:

```
ABCDEF
CDE
```

Extended versions of the byte-to-string constructors are also defined in which you can specify the character encoding that determines how bytes are converted to characters. However, you will often want to use the default encoding provided by the platform.

NOTE The contents of the array are copied whenever you create a **String** object from an array. If youmodify the contents of the array after you have created the string, the **String** will be unchanged.

You can construct a **String** from a **StringBuffer** by using the constructorshown here:

String(StringBuffer *strBufObj*)

You can construct a **String** from a **StringBuilder** by using this constructor:

String(StringBuilder *strBuildObj*)

The following constructor supports the extended Unicode character set:

String(int *codePoints*[ ], int *startIndex*, int *numChars*)

Here, *codePoints* is an array that contains Unicode code points. The resulting string is constructed from the range that begins at *startIndex* and runs for *numChars*.

There are also constructors that let you specify a **Charset**.

NOTE A discussion of Unicode code points and how they are handled by Java is found in Chapter 18.

String Length

The length of a string is the number of characters that it contains. To obtain this value, call the **length( )** method, shown here:

int length( )

The following fragment prints "3", since there are three characters in the string **s**:

```
char chars[] = { 'a', 'b', 'c' };
String s = new String(chars);
System.out.println(s.length());
```

String Concatenation

In general, Java does not allow operators to be applied to **String** objects. The one exception to this rule is the + operator, which concatenates two strings, producing a **String** object as the result. This allows you to chain together a series of + operations. For example, the following fragment concatenates three strings:

```
String age = "9";
String s = "He is " + age + " years old.";
System.out.println(s);
```

This displays the string "He is 9 years old."

One practical use of string concatenation is found when you are creating very long strings. Instead of letting long strings wrap around within your source code, you can break them into smaller pieces, using the + to concatenate them. Here is an example:

```
// Using concatenation to prevent long lines.
class ConCat {
  public static void main(String args[]) {
    String longStr = "This could have been " +
      "a very long line that would have " +
      "wrapped around.  But string concatenation " +
      "prevents this.";

    System.out.println(longStr);
  }
}
```

String Concatenation with Other Data Types

You can concatenate strings with other types of data. For example, consider this slightly different version of the earlier example:

```
int age = 9;
String s = "He is " + age + " years old.";
System.out.println(s);
```

In this case, **age** is an **int** rather than another **String**, but the output produced is the same as before. This is because the **int** value in **age** is automatically converted into its string representation within a **String** object. This string is then concatenated as before. The compiler will convert an operand to its string equivalent whenever the other operand of the **+** is an instance of **String**.

Be careful when you mix other types of operations with string concatenation expressions, however. You might get surprising results. Consider the following:

```
String s = "four: " + 2 + 2;
System.out.println(s);
```

This fragment displays

```
four:     22
```

rather than the

```
four: 4
```

that you probably expected. Here's why. Operator precedence causes the concatenation of "four" with the string equivalent of 2 to take place first. This result is then concatenated with the string equivalent of 2 a second time. To complete the integer addition first, you must use parentheses, like this:

```
String s = "four: " + (2 + 2);
```

Now **s** contains the string "four: 4".

String Comparison

The **String** class includes a number of methods that compare strings or substrings within strings. Several are examined here.

equals( ) and equalsIgnoreCase( )

To compare two strings for equality, use **equals( )**. It has this general form:

boolean equals(Object *str*)

Here, *str* is the **String** object being compared with the invoking **String** object. It returns **true** if the strings contain the same characters in the same order, and **false** otherwise. The comparison is case-sensitive.

To perform a comparison that ignores case differences, call

equalsIgnoreCase( ). When it compares two strings, it considers **A-Z** to be the same as **a-z**. It has this general form:

boolean equalsIgnoreCase(String *str*)

Here, *str* is the **String** object being compared with the invoking **String** object. It,

too, returns **true** if the strings contain the same characters in the same order, and **false** otherwise.

Here is an example that demonstrates **equals( )** and **equalsIgnoreCase( )**:

```
// Demonstrate equals() and equalsIgnoreCase().
class equalsDemo {
  public static void main(String args[]) {
    String s1 = "Hello";
    String s2 = "Hello";
    String s3 = "Good-bye";
    String s4 = "HELLO";
    System.out.println(s1 + " equals " + s2 + " -> " +
                       s1.equals(s2));
    System.out.println(s1 + " equals " + s3 + " -> " +
                       s1.equals(s3));
    System.out.println(s1 + " equals " + s4 + " -> " +
                       s1.equals(s4));
    System.out.println(s1 + " equalsIgnoreCase " + s4 + " -> " +
                       s1.equalsIgnoreCase(s4));
  }
}
```

The output from the program is shown here:

```
Hello  equals  Hello  ->  true
Hello equals Good-bye -> false
Hello equals HELLO -> false

Hello equalsIgnoreCase HELLO -> true
```

equals( ) Versus ==

It is important to understand that the **equals( )** method and the **==** operator perform two different operations. As just explained, the **equals( )** method compares the characters inside a **String** object. The **==** operator compares two object references to see whether they refer to the same instance. The following program shows how two different **String** objects can contain the same characters, but references to these objects will not compare as equal:

```
// equals() vs ==
class EqualsNotEqualTo {
  public static void main(String args[]) {
    String s1 = "Hello";
    String s2 = new String(s1);

    System.out.println(s1 + " equals " + s2 + " -> " +
                       s1.equals(s2));
    System.out.println(s1 + " == " + s2 + " -> " + (s1 == s2));
  }
}
```

The variable **s1** refers to the **String** instance created by **"Hello"**. The object referred to by **s2** is created with **s1** as an initializer. Thus, the contents of the two **String** objects are identical, but they are distinct objects. This means that **s1** and **s2** do not refer to the same objects and are, therefore, not ==, as is shown here by the output of the preceding example:

```
Hello equals Hello -> true
Hello == Hello -> false
```

compareTo( )

Often, it is not enough to simply know whether two strings are identical. For sorting applications, you need to know which is *less than*, *equal to*, or *greater than* the next. A string is less than another if it comes before the other in dictionary order. A string is greater than another if it comes after the other in dictionary order. The method **compareTo( )** serves this purpose. It is specified by the **Comparable<T>** interface, which **String** implements. It has this general form:

int compareTo(String *str*)

Here, *str* is the **String** being compared with the invoking **String**. The result of the comparison is returned and is interpreted as shown here:

Value	Meaning
Less than zero	The invoking string is less than *str*.
Greater than zero	The invoking string is greater than *str*.
Zero	The two strings are equal.

Here is a sample program that sorts an array of strings. The program uses **compareTo( )** to determine sort ordering for a bubble sort:

```
// A bubble sort for Strings.
class SortString {
  static String arr[] = {
    "Now", "is", "the", "time", "for", "all", "good", "men",
    "to", "come", "to", "the", "aid", "of", "their", "country"
  };
  public static void main(String args[]) {
    for(int j = 0; j < arr.length; j++) {
      for(int i = j + 1; i < arr.length; i++) {
        if(arr[i].compareTo(arr[j]) < 0) {
          String t = arr[j];
          arr[j] = arr[i];
          arr[i] = t;
        }
      }
      System.out.println(arr[j]);
    }
  }
}
```

The output of this program is the list of words:

```
Now aid
all
come
country
for
good is
```

```
men
of
the
the
their
time
to to
```

As you can see from the output of this example, **compareTo( )** takes into account uppercase and lowercase letters. The word "Now" came out before all the others because it begins with an uppercase letter, which means it has a lower value in the ASCII character set.

If you want to ignore case differences when comparing two strings, use **compareToIgnoreCase( )**, as shown here:

int compareToIgnoreCase(String *str*)

This method returns the same results as **compareTo( )**, except that case differences are ignored. You might want to try substituting it into the previous program. After doing so, "Now" will no longer be first.

StringBuffer

StringBuffer supports a modifiable string. As you know, **String** represents fixed-length, immutable character sequences. In contrast, **StringBuffer** represents growable and writable character sequences. **StringBuffer** may have characters and substrings inserted in the middle or appended to the end. **StringBuffer** will automatically grow to make room for such additions and often has more characters preallocated than are actually needed, to allow room for growth.

StringBuffer Constructors

StringBuffer defines these four constructors:

StringBuffer( )

StringBuffer(int *size*)
StringBuffer(String *str*)
StringBuffer(CharSequence *chars*)

The default constructor (the one with no parameters) reserves room for 16 characters without reallocation. The second version accepts an integer argument that explicitly sets the size of the buffer. The third version accepts a **String** argument that sets the initial contents of the **StringBuffer** object and reserves room for 16 more characters without reallocation. **StringBuffer** allocates room for 16 additional characters when no specific buffer length is requested, because reallocation is a costly process in terms of time. Also, frequent reallocations can fragment memory. By allocating room for a few extra characters, **StringBuffer** reduces the number of reallocations that take place. The fourth constructor creates an object that contains the character sequence contained in *chars* and reserves room for 16 more characters.

length( ) and capacity( )

The current length of a **StringBuffer** can be found via the **length( )** method, while the total allocated capacity can be found through the **capacity( )** method. They have the following general forms:

int length( )
int capacity( )

Here is an example:

```
// StringBuffer length vs. capacity.
class StringBufferDemo {
  public static void main(String args[]) {
    StringBuffer sb = new StringBuffer("Hello");

    System.out.println("buffer = " + sb);
    System.out.println("length = " + sb.length());
    System.out.println("capacity = " + sb.capacity());
  }
}
```

Here is the output of this program, which shows how **StringBuffer** reserves extra space for additional manipulations:

```
buffer = Hello
length = 5
capacity = 21
```

Since **sb** is initialized with the string "Hello" when it is created, its length is 5. Its capacity is 21 because room for 16 additional characters is automatically added.

ensureCapacity( )

If you want to preallocate room for a certain number of characters after a **StringBuffer** has been constructed, you can use **ensureCapacity( )** to set the size of the buffer. This is useful if you know in advance that you will be appending a large number of small strings to a **StringBuffer**. **ensureCapacity( )** has this general form:

void ensureCapacity(int *minCapacity*)

Here, *minCapacity* specifies the minimum size of the buffer. (A buffer larger than *minCapacity* may be allocated for reasons of efficiency.)

setLength( )

To set the length of the string within a **StringBuffer** object, use **setLength( )**. Its general form is shown here:

void setLength(int *len*)

Here, *len* specifies the length of the string. This value must be nonnegative.

When you increase the size of the string, null characters are added to the end. If you call **setLength( )** with a value less than the current value returned by **length( )**, then the characters stored beyond the new length will be lost. The **setCharAtDemo** sample program in the following section uses **setLength( )** to shorten a **StringBuffer**.

charAt( ) and setCharAt( )

The value of a single character can be obtained from a **StringBuffer** via the **charAt( )** method. You can set the value of a character within a **StringBuffer**using **setCharAt( )**. Their general forms are shown here:

char charAt(int *where*)

void setCharAt(int *where*, char *ch*)

For **charAt( )**, *where* specifies the index of the character being obtained. For **setCharAt( )**, *where* specifies the index of the character being set, and *ch* specifies the new value of that character. For both methods, *where* must be nonnegative and must not specify a location beyond the end of the string.

The following example demonstrates **charAt( )** and **setCharAt( )**:

```
// Demonstrate charAt() and setCharAt().
class setCharAtDemo {
  public static void main(String args[]) {
    StringBuffer sb = new StringBuffer("Hello");
    System.out.println("buffer before = " + sb);
    System.out.println("charAt(1) before = " + sb.charAt(1));

    sb.setCharAt(1, 'i');
    sb.setLength(2);
    System.out.println("buffer after = " + sb);
    System.out.println("charAt(1) after = " + sb.charAt(1));
  }
}
```

Here is the output generated by this program:

```
buffer before = Hello
charAt(1) before = e
buffer after = Hi
charAt(1) after = i
```

getChars( )

To copy a substring of a **StringBuffer** into an array, use the **getChars( )** method. It has this general form:

void getChars(int *sourceStart*, int *sourceEnd*, char *target*[ ], int *targetStart*)Here,

sourceStart specifies the index of the beginning of the substring, and

sourceEnd specifies an index that is one past the end of the desired substring. This means that the substring contains the characters from *sourceStart* through *sourceEnd*–1. The array that will receive the characters is specified by *target*.

The index within *target* at which the substring will be copied is passed in *targetStart*. Care must be taken to assure that the *target* array is large enough to hold the number of characters in the specified substring.

append( )

The **append( )** method concatenates the string representation of any other type of data to the end of the invoking **StringBuffer** object. It has several overloaded versions. Here are a few of its forms:

StringBuffer append(String *str*)
StringBuffer append(int *num*)
StringBuffer append(Object *obj*)

First, the string representation of each parameter is obtained. Then, the result is appended to the current **StringBuffer** object. The buffer itself is returned by each version of **append( )**. This allows subsequent calls to be chained together, as shown in the following example:

```
// Demonstrate append().
class appendDemo {
  public static void main(String args[]) {
    String s;
    int a = 42;
    StringBuffer sb = new StringBuffer(40);

    s = sb.append("a = ").append(a).append("!").toString();
    System.out.println(s);
  }
}
```

The output of this example is shown here:

```
a = 42!
```

insert( )

The **insert( )** method inserts one string into another. It is overloaded to accept values of all the primitive types, plus **String**s, **Object**s, and **CharSequence**s. Like **append( )**, it obtains the string representation of the value it is called with.

This string is then inserted into the invoking **StringBuffer** object. These are a few of its forms:

StringBuffer insert(int *index*, String *str*)
StringBuffer insert(int *index*, char *ch*)
StringBuffer insert(int *index*, Object *obj*)

Here, *index* specifies the index at which point the string will be inserted into the invoking **StringBuffer** object.

The following sample program inserts "like" between "I" and "Java":

```
// Demonstrate insert().
class insertDemo {
  public static void main(String args[]) {
    StringBuffer sb = new StringBuffer("I Java!");

    sb.insert(2, "like ");
    System.out.println(sb);
  }
}
```

The output of this example is shown here:

```
I like Java!
```

reverse( )

You can reverse the characters within a **StringBuffer** object using **reverse( )**,shown here:

StringBuffer reverse( )

This method returns the reverse of the object on which it was called. The following program demonstrates **reverse( )**:

```
// Using reverse() to reverse a StringBuffer.
class ReverseDemo {
  public static void main(String args[]) {
    StringBuffer s = new StringBuffer("abcdef");

    System.out.println(s);
    s.reverse();
    System.out.println(s);
  }
}
```

Here is the output produced by the program:

```
abcdef
fedcba
```

delete( ) and deleteCharAt( )

You can delete characters within a **StringBuffer** by using the methods **delete( )** and **deleteCharAt( )**. These methods are shown here:

StringBuffer delete(int *startIndex*, int *endIndex*)
StringBuffer deleteCharAt(int *loc*)

The **delete( )** method deletes a sequence of characters from the invoking object. Here, *startIndex* specifies the index of the first character to remove, and *endIndex* specifies an index one past the last character to remove. Thus, the substring deleted runs from *startIndex* to *endIndex*–1. The resulting **StringBuffer** object is returned.

The **deleteCharAt( )** method deletes the character at the index specified by *loc*. It returns the resulting **StringBuffer** object.

Here is a program that demonstrates the **delete( )** and **deleteCharAt( )** methods:

```
// Demonstrate delete() and deleteCharAt()
class deleteDemo {
  public static void main(String args[]) {
    StringBuffer sb = new StringBuffer("This is a test.");

    sb.delete(4, 7);
    System.out.println("After delete: " + sb);

    sb.deleteCharAt(0);
    System.out.println("After deleteCharAt: " + sb);
  }
}
```

The following output is produced:

```
After delete: This a test. After
deleteCharAt: his a test.
```

replace( )

You can replace one set of characters with another set inside a **StringBuffer** object by calling **replace( )**. Its signature is shown here:

StringBuffer replace(int *startIndex*, int *endIndex*, String *str*)

The substring being replaced is specified by the indexes *startIndex* and *endIndex*. Thus, the substring at *startIndex* through *endIndex*–1 is replaced. The replacement string is passed in *str*. The resulting **StringBuffer** object is returned.

The following program demonstrates **replace( )**:

```
// Demonstrate replace()
class replaceDemo {
  public static void main(String args[]) {
    StringBuffer sb = new StringBuffer("This is a test.");

    sb.replace(5, 7, "was");
    System.out.println("After replace: " + sb);
  }
}
```

Here is the output:

```
After replace: This was a test.
```

substring( )

You can obtain a portion of a **StringBuffer** by calling **substring( )**. It has the following two forms:

String substring(int *startIndex*)

String substring(int *startIndex*, int *endIndex*)

The first form returns the substring that starts at *startIndex* and runs to the end ofthe invoking **StringBuffer** object. The second form returns the substring that starts at *startIndex* and runs through *endIndex*–1. These methods work just likethose defined for **String** that were described earlier.

Java String compare

We can compare String in Java on the basis of content and reference.

There are three ways to compare String in Java:

1. By Using equals() Method
2. By Using == Operator
3. By compareTo() Method

1) By Using equals() Method

The String class equals() method compares the original content of the string. It compares values of string for equality. String class provides the following two methods:

- **public boolean equals(Object another)** compares this string to the specified object.
- **public boolean equalsIgnoreCase(String another)** compares this string to another string, ignoring case.

StringDemo.java

```java
public class StringDemo {
    public static void main(String args[]) {
        String s1 = "Sunil";
        String s2 = "Sunil";
        String s3 = new String("Sunil");
        String s4 = "Bist";
        System.out.println(s1.equals(s2));//true
        System.out.println(s1.equals(s3));//true
        System.out.println(s1.equals(s4));//false
    }
}
```

Output:

true
true
false

In the above code, two strings are compared using **equals()** method of **String** class. And the result is printed as boolean values, **true** or **false**.

StringDemo2.java

```java
public class StringDemo2{
    public static void main(String args[]){
        String s1="Sunil";
        String s2="SUNIL";

        System.out.println(s1.equals(s2));//false
        System.out.println(s1.equalsIgnoreCase(s2));//true
    }
}
```

Output:

false
true

In the above program, the methods of **String** class are used. The **equals()** method returns true if String objects are matching and both strings are of same case. **equalsIgnoreCase()** returns true regardless of cases of strings.

2) By Using == operator

The == operator compares references not values.

StringDemo3.java

```
public class StringDemo3{
    public static void main(String args[]){
        String s1="Sunil";
        String s2="Sunil";
        String s3=new String("Sunil");
        System.out.println(s1==s2);
        //true (because both refer to same instance)
        System.out.println(s1==s3);
        //false(because s3 refers to instance created in nonpool)
    }
}
```

The above code, demonstrates the use of == operator used for comparing two ***String*** objects.

3) By Using compareTo() method

The String class compareTo() method compares values lexicographically and returns an integer value that describes if first string is less than, equal to or greater than second string.

Suppose s1 and s2 are two String objects. If:

- **s1 == s2** : The method returns 0.
- **s1 > s2** : The method returns a positive value.
- **s1 < s2** : The method returns a negative value.

StringDemo4.java

```
public class StringDemo4{
    public static void main(String args[]){
        String s1="Sunil";
        String s2="Sunil";
        String s3="Ravi";
        System.out.println(s1.compareTo(s2));//0
        System.out.println(s1.compareTo(s3));//1(because s1>s3)
        System.out.println(s3.compareTo(s1));//-1(because s3 < s1 )
    }
}
```

2.7.4 String Buffer

StringBuffer sb = new StringBuffer();

```
sb.append("Hello");
sb.append(" ");
sb.append("world");
String message = sb.toString();
```

StringBuffer class is used to create mutable (modifiable) strings. The StringBuffer class in Java is the same as the String class except it is mutable i.e. it can be changed.

Important Constructors of StringBuffer class
StringBuffer(): creates an empty string buffer with an initial capacity of 16.
StringBuffer(String str): creates a string buffer with the specified string.
StringBuffer(int capacity): creates an empty string buffer with the specified capacity as length.

1. append() method

```
StringBuffer sb = new StringBuffer("Hello ");
    sb.append("Java"); // now original string is changed
    System.out.println(sb);
```

2. insert() method
The insert() method inserts the given string with this string at the given position.

Example:

```
 StringBuffer sb = new StringBuffer("Hello ");
    sb.insert(1, "Java");
    // Now original string is changed
    System.out.println(sb);
```

3. replace() method
The replace() method replaces the given string from the specified beginIndex and endIndex-1.

Example:

```
StringBuffer sb = new StringBuffer("Hello");
    sb.replace(1, 3, "Java");
    System.out.println(sb);
```

4. delete() method
The delete() method of the StringBuffer class deletes the string from the specified beginIndex to endIndex-1.

Example:

```
StringBuffer sb = new StringBuffer("Hello");
    sb.delete(1, 3);
    System.out.println(sb);
```

5. reverse() method

The reverse() method of the StringBuilder class reverses the current string.

Example:

```
StringBuffer sb = new StringBuffer("Hello");
    sb.reverse();
    System.out.println(sb);
```

6. capacity() method

The capacity() method of the StringBuffer class returns the current capacity of the buffer. The default capacity of the buffer is 16. If the number of characters increases from its current capacity, it increases the capacity by (oldcapacity*2)+2.

For instance, if your current capacity is 16, it will be (16*2)+2=34.

Example:

```
StringBuffer sb = new StringBuffer();
    System.out.println(sb.capacity()); // default 16
    sb.append("Hello");
    System.out.println(sb.capacity()); // now 16
    sb.append("java is my favourite language");
    System.out.println(sb.capacity());
    // Now (16*2)+2=34     i.e (oldcapacity*2)+2
```

2.8 Methods in Java

- 2.8.1 Predefined methods
- 2.8.2 User defined methods
- 2.8.3 Static methods